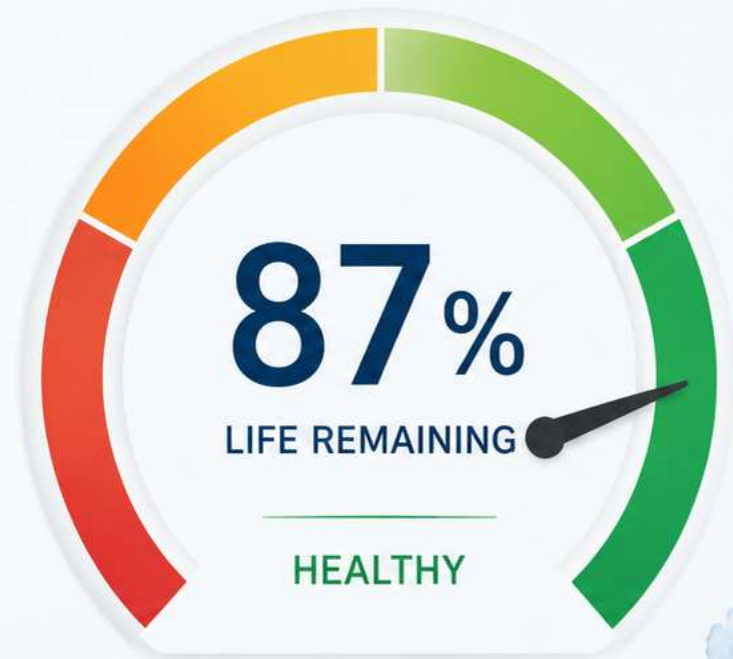


トランスの じゅみょう メーター Transformer Life Gauge

Real-Time Estimation of Transformer Insulation Life



Turning transformer thermal behavior into an easy-to-understand health indicator.

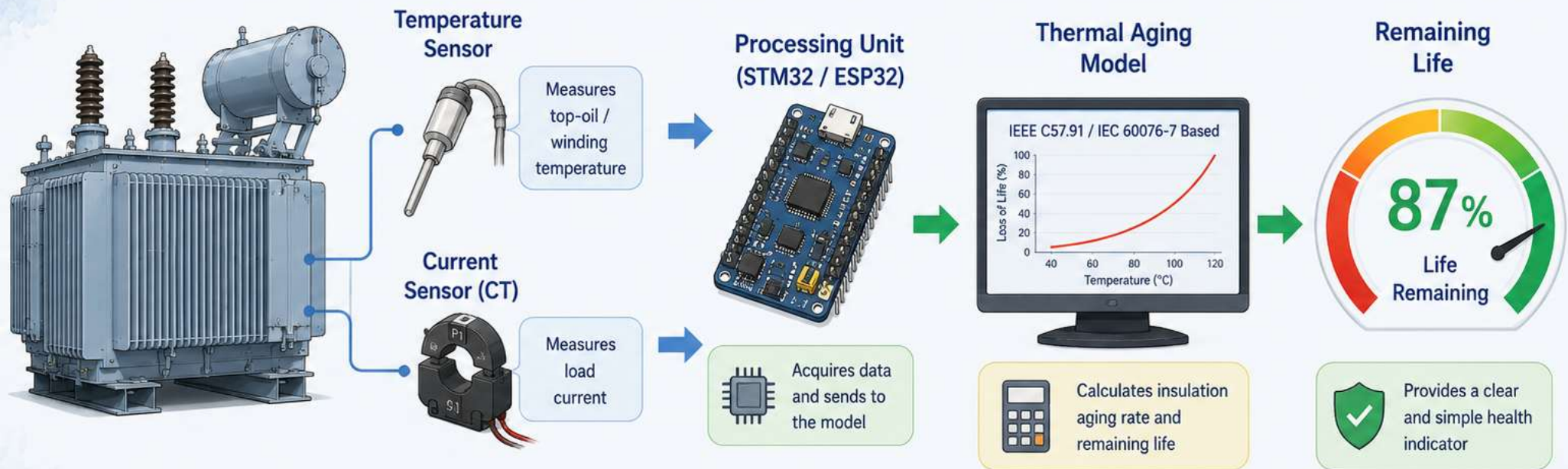
A presentation by
Rahul Sahu
Aditya University

トランスの じょうたい が わかる

あたらしい アイデア

Transformer Life Gauge Concept

Use key operating parameters and a thermal aging model to estimate remaining insulation life.



- ✓ Combines temperature and load current for accurate life estimation.
- ✓ Based on industry-standard thermal aging principles.
- ✓ Enables real-time assessment of transformer insulation health.



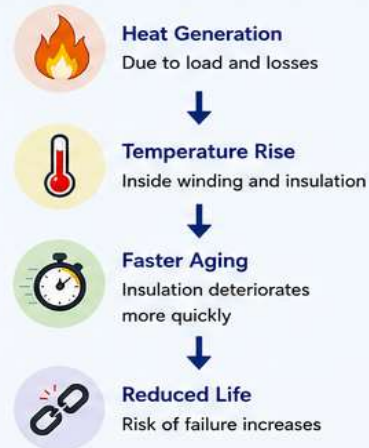
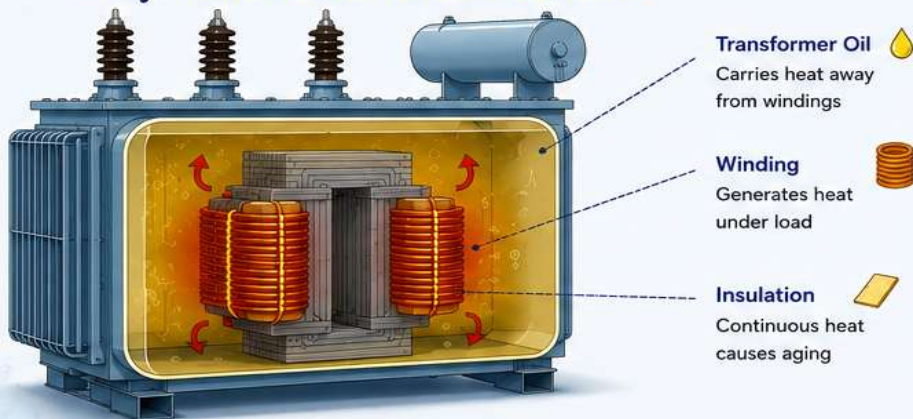
いまの ほしゅ と じゅみょうの わけ Why Current Maintenance Leads to Life Decrease

Maintenance is time-based, not condition-based. Heat is the main reason for insulation aging.



Maintenance happens at fixed intervals, not based on the real condition of the transformer.

Why Transformer Life Decreases



Industry Failure Data

Source: CIGRE Transformer Reliability Survey



25% of known failure causes come from insulation ageing + improper maintenance.

~0.3–0.4% / year
Overall large power transformer failure rate (CIGRE, modern units).



システムのこうせい Life Calculation Model



Three-step mathematical model to estimate transformer life in real-time.



MATLAB Offline Simulation (24-Hour Example)



Physics-Based Model
(IEEE C57.91)



Accurate Real-Time
Life Estimation



Smarter Loading
Decisions



Extends Transformer
Life & Reliability



Offline MATLAB Validation

The mathematical model is validated with a simulated daily load profile before deployment on the IoT hardware.

ライブ モニター Live Monitoring Dashboard

Real-time data visualization and instant alerts for better decision making.

モニター できるもの What We Monitor



おんど
Oil / Winding
Temperature



でんりゅう
Load Current



あぶらの じょうたい
Oil Quality



かんきょう
環境 条件
Environment



うんてん じかん
運転時間
Operating Hours

Transformer Life Gauge

Online



Overview



Trends



Alerts



Settings

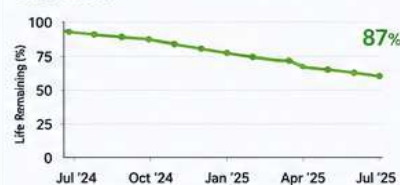
Life Remaining



Key Parameters

Oil Temperature	62.4 °C
Winding Temperature	68.7 °C
Load Current	412 A
Oil Quality	Good
Ambient Temperature	31.2 °C

Life Trend



Health Status



じょうたい は せいじょう です
Status is Normal

Last Updated: 12 May 2025, 10:30 AM

どこからでも
かんたんに チェック！

Easy to check
from anywhere!



ひとめで わかる
Easy to Understand
at a Glance



リアルタイム データ
Real-Time Data
Updates



アラートで まえに きづく
Early Alerts,
Prevent Failures



データで よりよい
かんりを サポート
Data-Driven
Better Management

まとめとこれから

Conclusion & Future Scope

Smarter monitoring today, stronger power tomorrow.

WHAT WE ACHIEVED



FUTURE SCOPE



あんぜんな データつうしん
Secure Data Transmission



リアルタイム モニター
Real-Time Monitoring



アラートで じこを よぼう
Early Fault Detection



データで じゅみようを のぼす
Data-Driven Life Extension



ありがとう
ございます

Thank You

Your time and attention are
highly appreciated.



Let's build a smarter
and safer power future,
together.



Reliable
Monitoring



Instant
Alerts



Smart
Insights



Better
Power Future